



DEPARTMENT OF MATHEMATICS
ACADEMIC YEAR: 2023-24 EVEN SEMESTER (BMATE201)
QUESTION BANK

MODULE-1: VECTOR CALCULUS

1. Find the unit vector normal to the surface $x^3y^3z^2 = 4$ at the point P (-1, -1, 2).
2. Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at (2, -1, 2).
3. Find the angle between the surfaces $xy^2z = 3x + z^2$ and $3x^2 - y^2 + 2z = 1$ at (1, -2, 1).
4. If $\phi = 2x^3y^2z^4$ find $\text{div}(\text{grad}\phi)$.
5. If $\vec{F} = (3xz - 3yz)\hat{i} + (3y^2 - 3xz)\hat{j} + (3z^3 - 3xy)\hat{k}$, find $\text{div}\vec{F}$.
6. Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at the point (2, -1, 2) in the direction of $2\hat{i} - 3\hat{j} + 6\hat{k}$.
7. Find the directional derivative of $\phi = x^2yz + 4xz^2$ at the point (1, -2, -1) in the direction of $2\hat{i} - \hat{j} - 2\hat{k}$.
8. Find $\text{div}\vec{F}$ and $\text{curl}\vec{F}$ of $\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$.
9. Find $\nabla\phi$ if $\phi = x^3 + y^3 + z^3 - 3xyz$ at the point (1, -1, 2).
10. Find div and curl of vector $\vec{v} = (xyz)\hat{i} + (3x^2y)\hat{j} + (xz^2 - y^2z)\hat{k}$ at the point (2, -1, 1).
11. Show that vector field $F = \frac{x\hat{i} + y\hat{j}}{x^2 + y^2}$ is both solenoidal and irrotational.
12. Find the value of a, b, c such that $\vec{F} = (x + y + az)\hat{i} + (bx + 2y - z)\hat{j} + (x + cy + 2z)\hat{k}$ is irrotational also find the scalar potential ϕ such that $\vec{F} = \nabla\phi$.
13. Show that $\vec{F} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is irrotational and find ϕ such that $\vec{F} = \nabla\phi$.
14. Show that $\vec{F} = (2xy + z^2)\hat{i} + (x^2 + 2yz)\hat{j} + (y^2 + 2xz)\hat{k}$ is irrotational and find ϕ such that $\vec{F} = \nabla\phi$.
15. If $\vec{F} = 2x^2\hat{i} - 3yz\hat{j} + xz^2\hat{k}$ and $\phi = 2z - x^3y$, find $\vec{F} \cdot \nabla\phi$ and $\vec{F} \times (\nabla\phi)$ at (1, -1, 1).
16. Find the constant a , such that the vector field $\vec{F} = 2xy^2z^2\hat{i} + 2x^2yz^2\hat{j} + ax^2y^2z\hat{k}$ is irrotational.
17. If $\phi(x, y, z) = xy^2 + yz^3$, find $\nabla\phi$ and $|\nabla\phi|$ at (1, -2, -1).

18. Find the value of a, b such that $\vec{F} = (axy + z^3)\hat{i} + (3x^3 - z)\hat{j} + (bxz^2 - y)\hat{k}$ is irrotational also find the scalar potential ϕ such that $\vec{F} = \nabla\phi$.
19. Using green's theorem evaluate $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is boundary of the region enclosed by $y = \sqrt{x}$ and $y = x^2$
20. If $\vec{F} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$ evaluate $\oint_C \vec{F} \cdot d\vec{r}$ from $(0,0,0)$ to $(1,1,1)$ along the curve $x = t, y = t^2, z = t^3$.
21. If $\vec{F} = xy\hat{i} + yz\hat{j} + zx\hat{k}$ evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where c is the curve represented by $x = t, y = t^2, z = t^3, -1 \leq t \leq 1$.
22. Verify green's theorem for evaluate $\oint (xy + y^2)dx + (x^2)dy$ where C is closed curve of the region bounded by $y = x$ and $y = x^2$
23. Verify stokes's theorem for the vector $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ taken round the rectangle bounded by $x = 0, x = a, y = 0$ and $y = b$.
24. Verify stokes's theorem for the vector $\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$ taken round the rectangle bounded by $x = \pm a, y = 0$ and $y = b$.
25. Evaluate $\oint (xy)dx + (xy^2)dy$ by stokes's theorem where C is the square in the XY plane with vertices $(1,0), (-1,0), (0,1), (0,-1)$